PROJECT ASSIGNMENT 2

Group10-PA2 User Analysis

FIFA.com and Chess.com

Course	CSC13112 - UI/UX Design, FIT-HCMUS
Group	Group10
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1. Analysis boundary and source discipline

Simulated brainstorming record: The workshop, raw-note generation, votes, discussion, attendance, and consensus below are scenario-based synthetic evidence prepared for course reporting. They are not contemporaneous meeting evidence.

Screen evidence: Visible interface states seed observable notes only.

Simulated study: Participant-behavior notes and votes are synthetic scenario records.

Design inference: Clusters and priorities are interpretations that require PA3 testing.

2. Simulated brainstorming session

Field	Record
Record type	Simulated brainstorming record
Date and duration	22 Jul 2026, 19:30-20:45 (75 minutes)
Participants	Le Minh; Nguyen Vu Bach; Pham Nguyen Gia Bao; Trang Minh Nhut
Objective	Transform PA1 continuity, screen evidence, and simulated-study patterns into prioritized PA2 problems.
Silent note generation	12 minutes; each member drafts atomic notes independently.
Round-robin sharing	16 minutes; one note per turn, no evaluation during sharing.
Clustering	18 minutes; group by user task and breakdown rather than feature.
Duplicate merging	8 minutes; retain the clearest atomic wording and all source tags.
Cluster naming	6 minutes; name clusters with user-facing task language.
Prioritization	8 minutes; six simulated votes per member per product.
Consensus	5 minutes; review weighted scores and resolve the selected problem statement.
Owner assignment	2 minutes; Bach owns FIFA synthesis, Bao owns Chess synthesis, Nhut owns diagrams/HCI mapping, Le owns integration/traceability.

3. Representative Notes

Fourteen representative notes remain in the main report. All 72 atomic notes are retained in Appendix A.

Note ID	Product	Cluster	Representative note
F-RN-01	FIFA	Ticket status clarity	Current sale state needs one label
F-RN-02	FIFA	Ticket status clarity	Missing state differs from sold out
F-RN-03	FIFA	Ticket status clarity	Resale must not look like primary sale
F-RN-04	FIFA	Ticket status clarity	Waiting room is a temporary state
F-RN-05	FIFA	Ticket status clarity	Hospitality is a distinct option
F-RN-06	FIFA	Ticket status clarity	State comparison belongs in one view
F-RN-07	FIFA	Handoff trust	Preview the destination domain
C-RN-01	Chess	Entry choice overload	Beginner review needs one named entry
C-RN-02	Chess	Entry choice overload	Analysis setup paths compete
C-RN-03	Chess	Entry choice overload	Feature names require recall
C-RN-04	Chess	Entry choice overload	Completed game should expose review
C-RN-05	Chess	Entry choice overload	Expert view opens too early
C-RN-06	Chess	Entry choice overload	Outcome labels are easier to recognize
C-RN-07	Chess	Learning priority	Show one important mistake first

4. Affinity diagrams and cluster synthesis

Prod uct	Cluster	Co unt	Representative notes	Pattern	Affected task	Sever ity	Design direction
FIFA	Ticket status clarity	6	Current sale state needs one label; Missing state differs from sold out	Repeated task-state or orientation breakdown	Compare ticket states	Critic al	One comparison dashboard
FIFA	Handoff trust	6	Preview the destination domain; Explain why departure is needed	Repeated task-state or orientation breakdown	Preview destination	High	Destination preview + stay/return
FIFA	Mobile overview	6	Keep selected tournament first; Avoid a long undifferentiated card stack	Repeated task-state or orientation breakdown	Resume on mobile	Mediu m	Persist tournament context
FIFA	Browse load	6	Ticket entry competes with content; Repeated cards increase scanning	Repeated task-state or orientation breakdown	Find ticket entry	Mediu m	Task-led entry
FIFA	Freshness and alerts	6	Show last-updated time; Name the governed source	Repeated task-state or orientation breakdown	Inspect freshness / subscribe	Mediu m	Timestamp + controlled alerts
FIFA	Recovery and	6	Retry missing state without	Repeated task-	Recover	Mediu	State-specific

Product	Cluster	Count	Representative notes	Pattern	Affected task	Severity	Design direction
	return		losing context; Network loss should retain the last snapshot	state or orientation breakdown	and return	m	recovery
Chess	Entry choice overload	6	Beginner review needs one named entry; Analysis setup paths compete	Repeated task-state or orientation breakdown	Open review	Critical	Beginner review preset
Chess	Learning priority	5	Show one important mistake first; Explain why before numeric depth	Repeated task-state or orientation breakdown	Identify mistake	High	One mistake first
Chess	Review setup	5	No game needs a clear setup route; Invalid PGN must preserve input	Repeated task-state or orientation breakdown	Open game	Medium	Validated setup states
Chess	Vocabulary load	5	Evaluation terms need inline help; Plain language should lead	Repeated task-state or orientation breakdown	Read explanation	High	Inline plain-language help
Chess	Mobile scanning	5	Keep the board and main mistake together; Collapse secondary panels	Repeated task-state or orientation breakdown	Review on mobile	Medium	Primary-card hierarchy
Ches	Practice	5	Link the mistake to one	Repeated task-	Continue	High	One relevant

Product	Cluster	Count	Representative notes	Pattern	Affected task	Severity	Design direction
s	continuation		lesson; Offer one relevant puzzle	state or orientation breakdown	practice		next exercise
Chess	Advanced depth disclosure	5	Reveal engine detail on demand; Do not remove expert access	Repeated task-state or orientation breakdown	Reveal detail	Medium	Progressive disclosure

5. Simulated dot voting

Simulated brainstorming record: Each member receives six votes for FIFA and six for Chess; totals equal 24 votes per product.

Member	Status	Handoff	Mobile	Browse	Freshness	Recovery	Total
Le Minh	3	2	0	0	1	0	6
Nguyen Vu Bach	3	2	0	0	0	1	6
Pham Nguyen Gia Bao	2	2	0	1	1	0	6
Trang Minh Nhut	2	2	1	0	1	0	6

FIFA cluster	Total votes
Ticket status clarity	10
Handoff trust	8
Freshness and alerts	3
Mobile overview	1

Member	Status	Handoff	Mobile	Browse	Freshness	Recovery	Total
Browse load				1			
Recovery and return				1			

Member	Entry	Priority	Setup	Vocabulary	Mobile	Practice	Depth	Total
Le Minh	1	0	0	2	0	2	1	6
Nguyen Vu Bach	1	1	1	1	0	1	1	6
Pham Nguyen Gia Bao	1	1	1	1	0	2	0	6
Trang Minh Nhut	1	1	1	1	1	0	1	6

Chess cluster	Total votes
Vocabulary load	5
Practice continuation	5
Entry choice overload	4
Learning priority	3
Review setup	3

<table>
<tr><th>Member</th><th>Status</th><th>Handoff</th><th>Mobile</th><th>Browse</th><th>Freshness</th><th>Recovery</th><th>Total</th></tr>
<tr><td colspan="4">Advanced depth disclosure</td><td colspan="4">3</td></tr>
<tr><td colspan="4">Mobile scanning</td><td colspan="4">1</td></tr>
<tr><th colspan="4">Field</th><th colspan="4">Record</th></tr>
<tr><td colspan="4">Discussion notes</td><td colspan="4">FIFA votes concentrate on status and handoff confidence. Chess votes split across vocabulary and continuation, while entry overload is the enabling problem.</td></tr>
<tr><td colspan="4">Tie-break rule</td><td colspan="4">If totals tie, select the item with higher severity, then stronger traceability to the PA3 prototype task.</td></tr>
<tr><td colspan="4">Consensus statement</td><td colspan="4">Prioritize FIFA ticket status plus handoff confidence, and treat Chess entry, explanation, and continuation as one guided beginner review problem.</td></tr>
<tr><td colspan="4">Selected problems</td><td colspan="4">FIFA: ticket status clarity and handoff trust.
Chess: guided beginner review.</td></tr>
</table>

6. Weighted prioritization

Formula: Weighted score = Severity x 25% + Task frequency x 15% + Risk x 20% + Reach x 10% + Evidence strength x 15% + PA3 relevance x 15%. Each factor is scored 1-5; maximum is 5.00.

6.1 FIFA candidates

Candidate	Criterion scores	Weighted	Score rationale
FIFA status + freshness	Severity 5 | Frequency 4 | Risk 5 | Reach 4 | Evidence 4 | PA3 5	4.60	Critical ambiguity; frequent planning; trust risk; broad reach; strong screen/synthetic trace; direct PA3 module.
FIFA handoff confidence	Severity 5 | Frequency 3 | Risk 5 | Reach 4 | Evidence 4 | PA3 5	4.45	Critical departure risk; lower frequency; broad consequence; direct PA3 decision.
FIFA mobile overview	Severity 3 | Frequency 4 | Risk 3 | Reach 4 | Evidence 4 | PA3 3	3.50	Frequent but lower consequence; evidence supports orientation rather than transaction risk.

6.2 Chess candidates

Candidate	Criterion scores	Weighted	Score rationale
Chess guided beginner review	Severity 5 | Frequency 5 | Risk 4 | Reach 5 | Evidence 4 | PA3 5	4.65	High learning severity/frequency/reach; strong PA3 relevance; synthetic outcomes require validation.
Chess mobile scanning	Severity 3 | Frequency 4 | Risk 3 | Reach 4 | Evidence 3 | PA3 3	3.35	Broad but secondary to entry and learning-priority breakdown.
Chess advanced depth	Severity 3 | Frequency 2 | Risk 3 | Reach 3 | Evidence 3 | PA3 4	3.00	Important control but lower beginner frequency; retained as a disclosure requirement.

7. Tough problems

7.1 TP-FIFA

TP-FIFA: Users need to compare current ticket state, freshness and official destination before leaving FIFA.com.

Field	Record
Affected users	Trust-sensitive planners and tournament followers
Context	Ticket planning before an outbound handoff
Screen evidence	F2-E03/E04/E09
Simulated study result	F-SIM-02/04/05/06 show status or handoff hesitation and 2-3/5 confidence
Root cause	State, freshness, and destination are separated
Consequence	Delayed action, unsafe assumption, or abandonment
Market gap	Official sports sites rarely combine governed state, freshness, and destination preview in one decision view
Scope	Comparison, preview, alerts, and recovery inside FIFA.com
Out of scope	Transaction, inventory creation, pricing, or partner redesign
PA3 test question	Can users accurately choose buy, wait, register interest, resale, hospitality, or stay?

7.2 TP-CHESS

TP-CHESS: Beginners need one interpretable mistake and one next practice action before advanced analysis.

Field	Record
Affected users	First-time beginners and returning low-analysis learners
Context	After-game review and practice continuation
Screen evidence	C2-E05/E07-E10
Simulated study result	C-SIM-01/02/04/05/06 show wrong paths, vocabulary hesitation, and continuation need
Root cause	Feature/setup organization precedes the learning outcome
Consequence	Expert view without an actionable learning priority
Market gap	Many analysis tools optimize depth before beginner interpretation and continuation
Scope	Beginner preset, one mistake, try move, explanation, practice, optional depth
Out of scope	Engine redesign, coaching guarantees, or removal of expert controls
PA3 test question	Can users explain one mistake and select one relevant practice action?

8. Design rationale and validation target

HCI rationale: Course basis: LN01 - Introduction - v2.pdf, pp. 20, 22, 24; LN02 - Fundamental Concepts - Usability Dimensions_2.pdf, pp. 62-67, 74-76, 83-84, 101-115; LN03 - UI Design Process.pdf, pp. 16-19, 23-24, 28-30, 34-36; LN04 - Task Analysis.pdf, pp. 3, 6-15, 39-49.

Design inference: The selection balances effectiveness, efficiency, learnability, memorability, errors, satisfaction, status visibility, recognition, progressive disclosure, user control, and error prevention.

8.1 Complete Affinity Diagram

FIFA Affinity Diagram - Part 1

Ticket status clarity

- One current state label
- Missing is not sold out
- Show last update

Pattern	Affected task	Severity	Design direction
State is compared before action	Compare ticket routes	Critical	Place state, freshness, and action together

Handoff trust

- Preview official domain
- Explain account change
- Keep a return route

Pattern	Affected task	Severity	Design direction
Leaving FIFA.com is a trust decision	Preview destination	High	Show domain, purpose, account, and return

Mobile overview

- Cards stack deeply
- Selected event disappears
- Filters require recall

Pattern	Affected task	Severity	Design direction
Mobile scanning loses context	Resume ticket planning	High	Persist a compact event and state summary

Figure UA-AFFINITY-FIFA-A. FIFA clusters, part 1: status clarity, handoff trust, and mobile overview.

FIFA Affinity Diagram - Part 2

Browse load

Many tournament cards	Routes are separated	Comparison needs backtracking

Pattern	Affected task	Severity	Design direction
Browsing creates memory work	Choose a ticket route	Medium	Group comparable routes in one decision view

Freshness and alerts

State changes over time	Timestamp is secondary	Alerts need consent

Pattern	Affected task	Severity	Design direction
Time affects action confidence	Wait or subscribe	High	Show freshness and permission-safe alerts

Recovery and return

Partner can fail	Network can fail	Valid answers should remain

Pattern	Affected task	Severity	Design direction
Failure should not reset planning	Recover from interruption	High	Preserve context and return to a safe step

Figure UA-AFFINITY-FIFA-B. FIFA clusters, part 2: browse load, freshness and alerts, and recovery and return.

Chess Affinity Diagram - Part 1

Entry choice overload

- Several analysis routes
- Names require recall
- Recent game action is apart

Pattern
Beginners choose the wrong entry

Affected task
Start game review

Severity
High

Design direction
Add Beginner Review beside the game result

Learning priority

- Many findings compete
- First mistake is unclear
- Next task is not ranked

Pattern
The learner lacks one priority

Affected task
Identify a mistake

Severity
High

Design direction
Show one important mistake first

Review setup

- No recent game
- Import can fail
- PGN may be invalid

Pattern
Setup blocks the learning task

Affected task
Open a reviewable game

Severity
High

Design direction
Offer recent/import paths with validation

Figure UA-AFFINITY-CHESS-A. Chess clusters, part 1: entry choice, learning priority, review setup, and vocabulary load.

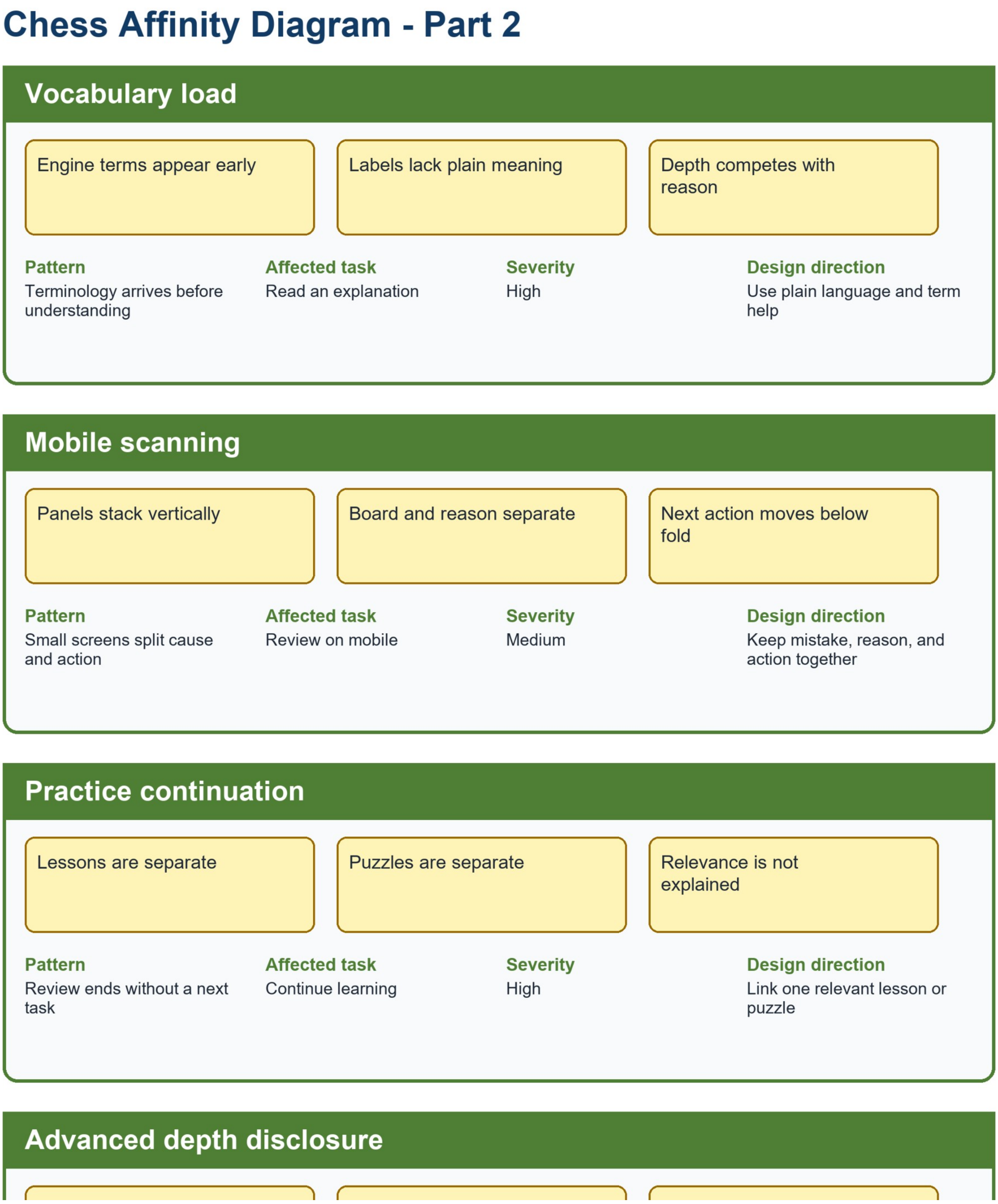

Figure UA-AFFINITY-CHESS-B. Chess clusters, part 2: mobile scanning, practice continuation, and advanced-depth disclosure.

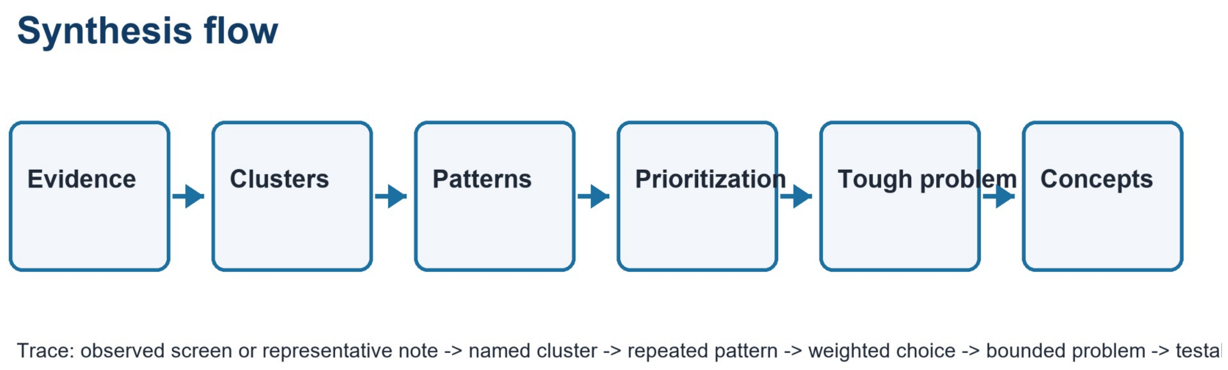

Figure UA-AF-SYNTHESIS. Evidence -> clusters -> patterns -> prioritization -> tough problem -> concepts.

Synthesis flow: Evidence -> Clusters -> Patterns -> Prioritization -> Tough problem -> Concepts. This sequence keeps the affinity diagram separate from a task sequence.

8.2 Voting Visualization

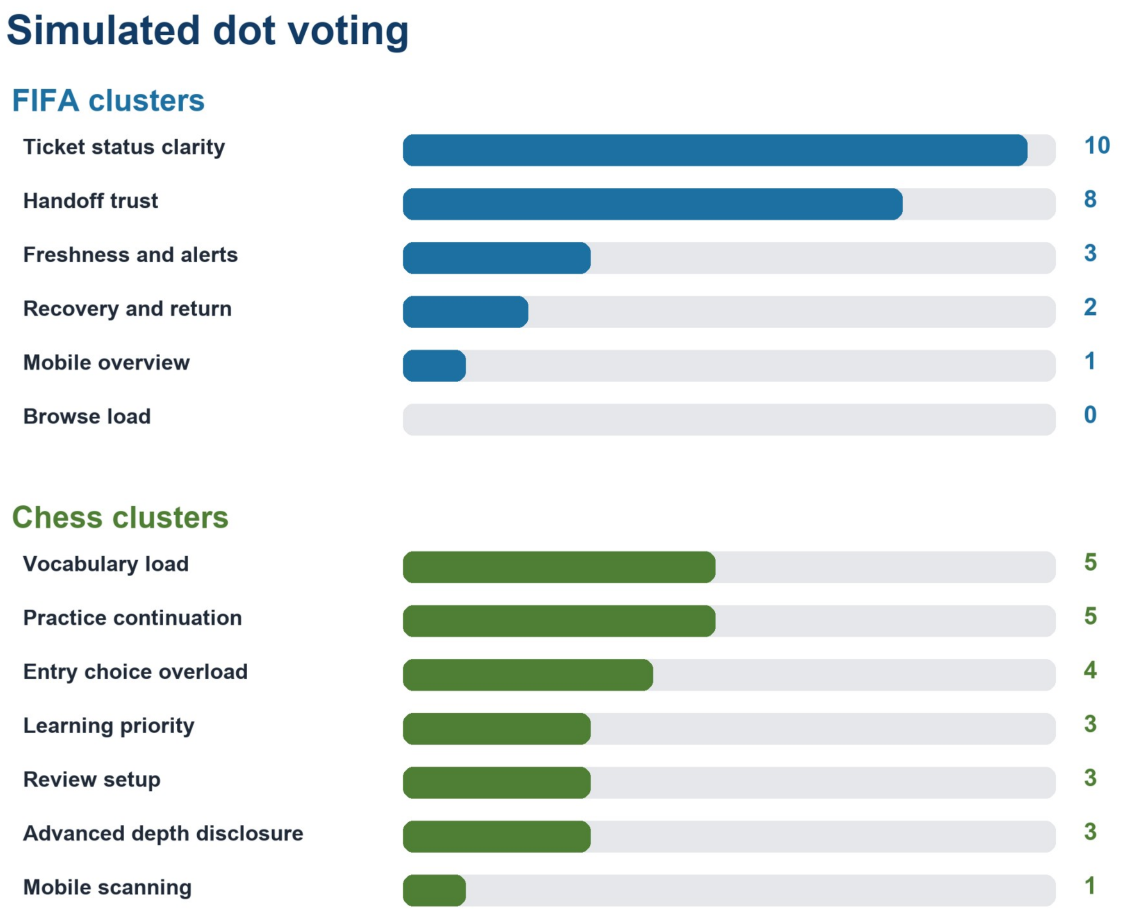

Figure UA-VOTE-01. Horizontal bars show the simulated dot counts; the tables retain member-level totals and tie-break detail.

8.3 Final Rationale

Final rationale: FIFA status and freshness remains the highest-priority problem because it combines high severity, trust risk, repeated planning use, and direct PA3 testability. Chess beginner review remains the selected problem because explanation and practice continuation jointly determine whether review becomes learning.

9. Practical Difficulties and Lessons Learned

Practical difficulty	How the group handled it	Lesson learned	Impact on the final version
Seventy-two notes could not remain legible on one page.	The group split product views and kept only three representative notes in each visible cluster.	Affinity synthesis should show the transformation, not only the notes.	The final figures expose pattern, task, severity, and design implication.
Use-case and PA1 terms drifted during clustering.	Stable FIFA and Chess cluster names were used across analysis and proposal sections.	Terminology is part of traceability.	The revised diagrams link each problem to one named design direction.

Appendix A - All 72 Raw Notes

Note ID	Product	Cluster	Atomic note	Source type
F-RN-01	FIFA	Ticket status clarity	Current sale state needs one label	screen evidence
F-RN-02	FIFA	Ticket status clarity	Missing state differs from sold out	simulated participant behavior
F-RN-03	FIFA	Ticket status clarity	Resale must not look like primary sale	PA1 inherited finding
F-RN-04	FIFA	Ticket status clarity	Waiting room is a temporary state	design inference
F-RN-05	FIFA	Ticket status clarity	Hospitality is a distinct option	screen evidence
F-RN-06	FIFA	Ticket status clarity	State comparison belongs in one view	simulated participant behavior
F-RN-07	FIFA	Handoff trust	Preview the destination domain	PA1 inherited finding
F-RN-08	FIFA	Handoff trust	Explain why departure is needed	design inference
F-RN-09	FIFA	Handoff trust	Offer stay on FIFA.com	screen evidence
F-	FIFA	Handoff trust	Preserve a safe return route	simulated

Note ID	Product	Cluster	Atomic note	Source type
RN-10				participant behavior
F-RN-11	FIFA	Handoff trust	Do not imply partner transaction status	PA1 inherited finding
F-RN-12	FIFA	Handoff trust	Keep tournament context through handoff	design inference
F-RN-13	FIFA	Mobile overview	Keep selected tournament first	screen evidence
F-RN-14	FIFA	Mobile overview	Avoid a long undifferentiated card stack	simulated participant behavior
F-RN-15	FIFA	Mobile overview	Restore filters on return	PA1 inherited finding
F-RN-16	FIFA	Mobile overview	Use labels that survive small screens	design inference
F-RN-17	FIFA	Mobile overview	Make status scannable without color	screen evidence
F-RN-18	FIFA	Mobile overview	Resume with one recognition step	simulated participant behavior
F-	FIFA	Browse load	Ticket entry competes with content	PA1 inherited

Note ID	Product	Cluster	Atomic note	Source type
RN-19				finding
F-RN-20	FIFA	Browse load	Repeated cards increase scanning	design inference
F-RN-21	FIFA	Browse load	Tournament identity should anchor actions	screen evidence
F-RN-22	FIFA	Browse load	Reduce cross-page comparison	simulated participant behavior
F-RN-23	FIFA	Browse load	Keep action wording consistent	PA1 inherited finding
F-RN-24	FIFA	Browse load	Separate browse from purchase intent	design inference
F-RN-25	FIFA	Freshness and alerts	Show last-updated time	screen evidence
F-RN-26	FIFA	Freshness and alerts	Name the governed source	simulated participant behavior
F-RN-27	FIFA	Freshness and alerts	Label stale data explicitly	PA1 inherited finding
F-RN-28	FIFA	Freshness and alerts	Allow relevant change alerts	design inference

Note ID	Product	Cluster	Atomic note	Source type
F-RN-29	FIFA	Freshness and alerts	Explain notification permission	screen evidence
F-RN-30	FIFA	Freshness and alerts	Let users edit or stop alerts	simulated participant behavior
F-RN-31	FIFA	Recovery and return	Retry missing state without losing context	PA1 inherited finding
F-RN-32	FIFA	Recovery and return	Network loss should retain the last snapshot	design inference
F-RN-33	FIFA	Recovery and return	Login mismatch needs safe recovery	screen evidence
F-RN-34	FIFA	Recovery and return	Expired return links need explanation	simulated participant behavior
F-RN-35	FIFA	Recovery and return	Partner failure must offer stay	PA1 inherited finding
F-RN-36	FIFA	Recovery and return	Never infer purchase completion	design inference
C-RN-01	Chess	Entry choice overload	Beginner review needs one named entry	screen evidence
C-	Ches	Entry choice	Analysis setup paths compete	simulated

Note ID	Product	Cluster	Atomic note	Source type
RN-02	s	overload		participant behavior
C-RN-03	Chess	Entry choice overload	Feature names require recall	PA1 inherited finding
C-RN-04	Chess	Entry choice overload	Completed game should expose review	design inference
C-RN-05	Chess	Entry choice overload	Expert view opens too early	screen evidence
C-RN-06	Chess	Entry choice overload	Outcome labels are easier to recognize	simulated participant behavior
C-RN-07	Chess	Learning priority	Show one important mistake first	PA1 inherited finding
C-RN-08	Chess	Learning priority	Explain why before numeric depth	design inference
C-RN-09	Chess	Learning priority	Make the next action explicit	screen evidence
C-RN-10	Chess	Learning priority	Keep the lesson goal visible	simulated participant behavior
C-	Ches	Learning priority	Do not present every candidate equally	PA1 inherited

Note ID	Product	Cluster	Atomic note	Source type
RN-11	s			finding
C-RN-12	Chess	Review setup	No game needs a clear setup route	design inference
C-RN-13	Chess	Review setup	Invalid PGN must preserve input	screen evidence
C-RN-14	Chess	Review setup	Board orientation needs confirmation	simulated participant behavior
C-RN-15	Chess	Review setup	Engine failure needs a safe fallback	PA1 inherited finding
C-RN-16	Chess	Review setup	Premium restriction needs honest scope	design inference
C-RN-17	Chess	Vocabulary load	Evaluation terms need inline help	screen evidence
C-RN-18	Chess	Vocabulary load	Plain language should lead	simulated participant behavior
C-RN-19	Chess	Vocabulary load	Terminology help should not interrupt	PA1 inherited finding
C-RN-20	Chess	Vocabulary load	Returning learners forget feature names	design inference

Note ID	Product	Cluster	Atomic note	Source type
C-RN-21	Chess	Vocabulary load	Recognition should replace recall	screen evidence
C-RN-22	Chess	Mobile scanning	Keep the board and main mistake together	simulated participant behavior
C-RN-23	Chess	Mobile scanning	Collapse secondary panels	PA1 inherited finding
C-RN-24	Chess	Mobile scanning	Use color-independent indicators	design inference
C-RN-25	Chess	Mobile scanning	Preserve state after interruption	screen evidence
C-RN-26	Chess	Mobile scanning	Maintain readable tap targets	simulated participant behavior
C-RN-27	Chess	Practice continuation	Link the mistake to one lesson	PA1 inherited finding
C-RN-28	Chess	Practice continuation	Offer one relevant puzzle	design inference
C-RN-29	Chess	Practice continuation	Preserve the reviewed position	screen evidence
C-	Ches	Practice	Let users continue later	simulated

Note ID	Product	Cluster	Atomic note	Source type
RN-30	s	continuation		participant behavior
C-RN-31	Chess	Practice continuation	Explain why the practice is relevant	PA1 inherited finding
C-RN-32	Chess	Advanced depth disclosure	Reveal engine detail on demand	design inference
C-RN-33	Chess	Advanced depth disclosure	Do not remove expert access	screen evidence
C-RN-34	Chess	Advanced depth disclosure	Remember the disclosure choice	simulated participant behavior
C-RN-35	Chess	Advanced depth disclosure	Keep beginner explanation stable	PA1 inherited finding
C-RN-36	Chess	Advanced depth disclosure	Separate learning priority from depth	design inference

Scenario-based synthetic evidence: All 72 notes are one-idea records; simulated participant behavior notes are not real observation claims.